

TASK -1

Relay stays **ON for 5 seconds** and **OFF for 2 seconds**.

MicroPython Code

```
import machine
import time

# 1. Configuration
RELAY_PIN = 15  # Change this to your actual GPIO pin number

# Set this to True if your relay is Active-High
# Set this to False if your relay is Active-Low (most common for
# Arduino/Pi modules)
ACTIVE_HIGH = True

# 2. Setup
relay = machine.Pin(RELAY_PIN, machine.Pin.OUT)

# Helper function to handle active-high vs active-low logic
def set_relay(state):
    if ACTIVE_HIGH:
        relay.value(1 if state == "ON" else 0)
    else:
        relay.value(0 if state == "ON" else 1)

# 3. Main Loop
print("Starting relay test (5s ON, 2s OFF). Press Ctrl+C to stop.")

try:
    while True:
        print("Relay is ON...")
        set_relay("ON")
        time.sleep(5)  # Keep ON for 5 seconds

        print("Relay is OFF...")
        set_relay("OFF")
        time.sleep(2)  # Keep OFF for 2 seconds

except KeyboardInterrupt:
    # Safe cleanup on exit
    set_relay("OFF")
    print("\nTest stopped. Relay turned off safely.")
```

NOTE :

If it stays ON for 2 seconds and OFF for 5 seconds: Your relay module is Active-Low. Simply change line 8 to `ACTIVE_HIGH = False`, re-run the script, and it will fix the timing automatically!